* Syllabus and Structure *

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* Engineering mathematics - II (107008)

* Course Contents:-

* Unit I - First Order ordinary diff. eqns
  * Exact Differential Equations
  * Equations reducible to exact forms
  * Linear Differential Equations
  * Equations reducible to linear form
  * Bernoulli's Equations.

* Unit II - Applications of Differential Equations
  * Orthogonal Trajectories.
  * Newton's law of cooling.
  * Kirchoff's law of electrical circuits.
  * Rectilinear motion.
  * Simple Harmonic Motion.
  * One dimensional Conduction of Heat.

* Unit - III - Integral Calculus
  * Reduction formulae
  * Beta and Gamma functions.
  * Differentiation Under Integral sign
  * Error functions.

* Unit - IV - Curve Tracing
  * Tracing of curves - Cartesian, Polar and Parametric curves
  * Rectification of curves

* Unit - V - Solid Geometry
  * Cartesian, spherical polar and Cylindrical coordinate systems

* sphere cone and cylinder.

* Unit VI :- Multiple Integrals and their Application
    * Double and Triple Integration
    * change of order of integration
    * Applications to find - Area, Volume, Mass, centre of gravity and Moment of Inertia.

* Text Books :-
  1. Higher Engineering Mathematics By B.V. Ramana (Tata M.C. Graw Hill)
  2. Higher Engineering Mathematics By B.S. Grewal. (Khanna Publications Delhi)

* Reference Books :-
  1. Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Eastern Ltd.)
  2. Applied Mathematics (Vol. I and II) By P.N. Wartikar and J.N. Wartikar Vidyarthi Griha Prakashan, Pune.

Unit I - Differential Equations

Chapter - I

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* Differential Equations :-

* Equation: An expression which contains L.H.S. and R.H.S. is called as an equation.

ex) $x^2+y^2 = 2xy$

* An equation containing the dependant variable, the independant variable and the derivatives or differential coefficients of variaous orders is called as Differential Equations.

* Order :- The order of D.E. is the order of the highest derivative appearing in the eqn.

Ex) $\frac{dy}{dx} \longrightarrow$ order: 1

$\frac{d^2y}{dx^2} \longrightarrow$ order: 2

$\frac{d^3y}{dx^2} \longrightarrow$ order: 3

* Degree :- The degree of D.E. is the power or exponent or degree of highest order derivative provided that is free from radical and fractions.

Ex) $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 = \frac{dy}{dx}$

order: 2 and degree 3.

Que: State the order and degree of the following differential Equations.

① $\sqrt{1+\frac{dy}{dx}} = \frac{d^2y}{dx^2} \longrightarrow$ ①

on squaring both sides

$1+\frac{dy}{dx} = \left(\frac{d^2y}{dx^2}\right)^2$

$\therefore$ order = 2 and degree = 2

Ex) $\sqrt[3]{\frac{d^2y}{dx^2}} = \sqrt{\frac{dy}{dx}}$

$\Rightarrow$ Take the cube of both sides

$$\frac{d^2y}{dx^2} = \left(\sqrt{\frac{dy}{dx}}\right)^3$$

Take the square of both sides

$$\left(\frac{d^2y}{dx^2}\right)^2 = \left(\frac{dy}{dx}\right)^3$$

$\therefore$ order = 2 and degree = 2.

* Formation of differential equations :
The process of differentiating an algebraic eqn to form D.E. which is free from arbitray constants is called as formation of DE.

* Pure Constants :- Ex) 3, 3 is a pure constant and it's value is 3.

* Arbritary Constant :- Ex) A, C, B,
Here A, C and B are arbitray constants and their values can vary from example to example.

* Note :- If there are n arbitrary constants then we will differentiate n times to form DE.

Ex ① Form the differential Equation whose general solution is $y = A\cos(nx) + B\sin(nx)$.

$\Rightarrow$ $y = A\cos(nx) + B\sin(nx)$. $\longrightarrow$ ①

Here, there are two arbitrary constants (i.e. A & B)
$\therefore$ We can differentiate only two times.

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Differentiate eqⁿ ① w.r.t. x

$$\frac{dy}{dx} = A\frac{d}{dx}(\cos nx) + B\frac{d}{dx}(\sin nx).$$

$$\frac{dy}{dx} = -nA\sin nx + nB\cos nx \longrightarrow ②.$$

$$\because \frac{d}{dx}(\cos nx) = -n\sin(nx)$$

$$\frac{d}{dx}(\sin nx) = n\cos(nx).$$

Again differentiate eqⁿ ② w.r.t. x.

$$\frac{d^2y}{dx^2} = -n^2A\cos nx - n^2B\sin nx$$

$$= -n^2\underbrace{(A\cos nx + B\sin nx)}_{y}$$

$$= -n^2y$$

$$\therefore \frac{d^2y}{dx^2} + n^2y = 0$$

order = 2
degree = 1.

Que: Form the differential Equation whose general solution is $y = c^2 + \frac{c}{x}$

⇒ Given, $y = c^2 + \frac{c}{x} \longrightarrow ①$

There is only one arbitrary constant (c), Hence we can differentiate only once.

diff eqⁿ ① w.r.t. x

$$\frac{dy}{dx} = 0 + c\left(\frac{-1}{x^2}\right)$$

$$\frac{dy}{dx} = \frac{-c}{x^2}$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{-1}{x^2}$$

$$c = -x^2\frac{dy}{dx} \longrightarrow ②$$

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put this value of c in eq^n ①

$$y = \left(-x^2 \frac{dy}{dx}\right)^2 + \left(\frac{-x^2\frac{dy}{dx}}{x}\right)$$

$$y = x^4\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx}$$

$$\therefore x^4\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} - y = 0$$

order = 1
degree = 2

H.W. ① Form DE whose general solution is

$$y = e^x(A\cos x + B\sin x).$$

$$\Rightarrow \frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0, \quad \text{order} = 2, \ \text{degree} = 1$$

* Differential Equation of First order and First degree:-

The D.E. of first order and first degree has form $Mdx + Ndy = 0$

$$M(x,y)\,dx + N(x,y)\,dy = 0.$$

or

$$\frac{dy}{dx} = f(x,y).$$

Ex) $\dfrac{dy}{dx} = \dfrac{x-3y}{2y-x}.$

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* Exact Differential Equations:-

A D.E. of the type $Mdx + Ndy = 0$ is called an exact D.E. iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

If D.E. is exact then the general solution is

$$\int_{y=const.} M\,dx + \int \begin{pmatrix}\text{terms of } N \text{ not} \\ \text{containing } x\end{pmatrix} dy = c$$

or

$$\int_{x=const.} N\,dy + \int \begin{pmatrix}\text{The terms of } M \\ \text{not containing } y\end{pmatrix} dx = c$$

* Flow chart:-

D.E. of 1st order and 1st degree

$Mdx + Ndy = 0$

↓

IF $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ then DE is exact

↓

Solution of D.E. is

$$\int_{y=const.} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$$

Que: Solve $(2xy^4 + \sin y)\,dx + (4x^2y^3 + x\cos y)\,dy = 0$

⇒ Given: $(2xy^4 + \sin y)\,dx + (4x^2y^3 + x\cos y)\,dy = 0 \rightarrow (1)$

compare with $Mdx + Ndy = 0$

$M = 2xy^4 + \sin y$ and $N = 4x^2y^3 + x\cos y$

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$M = 2xy^4 + \sin y$

Differentiate w.r.t. y partially

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(2xy^4 + \sin y)$$

$$= 2x\frac{\partial}{\partial y}(y^4) + \frac{\partial}{\partial y}(\sin y)$$

$$= 2x(4y^3) + \cos y$$

$$\boxed{\frac{\partial M}{\partial y} = 8xy^3 + \cos y} \longrightarrow (2)$$

~~M~~ $N = 4x^2y^3 + x\cos y$

Differentiating w.r.t. x partially

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(4x^2y^3 + x\cos y)$$

$$= 4y^3\frac{\partial}{\partial x}(x^2) + \cos y\frac{\partial}{\partial x}(x)$$

$$= 4y^3(2x) + \cos y(1)$$

$$\boxed{\frac{\partial N}{\partial x} = 8xy^3 + \cos y} \longrightarrow (3)$$

from (2) and (3)

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 8xy^3 + \cos y$$

$\therefore$ Given DE is exact

It's general solution is given by

$$\int_{y=\text{const.}} M\,dx + \int\left(\begin{array}{c}\text{terms of N not}\\ \text{containing } x\end{array}\right)dy = c$$

$$\int_{y=\text{const.}} (2xy^4 + \sin y)\,dx + \int(0)\,dy = c$$

$$2y^4\int x\,dx + \sin y\int dx = c$$

$\cancel{2}y^4\left(\frac{x^2}{\cancel{2}}\right) + \sin y\,(x) = C$

$$\boxed{x^2y^4 + x\sin y = C}$$

which is the required solution.

Ex) Solve $(y^2e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$

$\Rightarrow$ Given,

$(y^2e^{xy^2} + 4x^3)\,dx + (2xye^{xy^2} - 3y^2)\,dy = 0 \longrightarrow (1)$

compare with $Mdx + Ndy = 0$.

$M = y^2e^{xy^2} + 4x^3$ and $N = 2xye^{xy^2} - 3y^2$

diff. w.r.t. y partially ~~diff. w.r.t. x partially~~

$\frac{\partial M}{\partial y} = y^2e^{xy^2}(2xy) + e^{xy^2}(2y) + 0$ $\cancel{\frac{\partial N}{\partial x} =}$

$\frac{\partial M}{\partial y} = 2xy^3e^{xy^2} + 2ye^{xy^2} \longrightarrow (2)$

Now $N = 2xye^{xy^2} - 3y^2$

diff. w.r.t. x partially

$\frac{\partial N}{\partial x} = 2y\frac{\partial}{\partial x}(xe^{xy^2}) - \frac{\partial}{\partial x}(3y^2)$

$\frac{\partial N}{\partial x} = 2y\left[x.e^{xy^2}(y^2) + e^{xy^2}(1)\right] - 0$

$\frac{\partial N}{\partial x} = 2xy^3e^{xy^2} + 2ye^{xy^2} \longrightarrow (3)$

from (2) and (3)

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2xy^3e^{xy^2} + 2ye^{xy^2}$.

$\therefore$ The given D.E. is exact.

The general solution is given by

$\int\limits_{(y=\text{const})} M\,dx + \int \left(\begin{matrix}\text{The terms of } N \\ \text{not containing } x\end{matrix}\right) dy = C$

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$$\int_{(y=const)} \left(y^2 e^{xy^2} + 4x^3\right) dx + \int \left(-3y^2\right) dy = C$$

$$\therefore y^2 \int e^{xy^2} dx + 4\int x^3 dx - 3\int y^2 dy = C$$

$$\cancel{y^2}\left(\frac{e^{xy^2}}{\cancel{y^2}}\right) + \cancel{4}\left(\frac{x^4}{\cancel{4}}\right) - \cancel{3}\left(\frac{y^3}{\cancel{3}}\right) = C$$

$$\boxed{e^{xy^2} + x^4 - y^3 = C}$$

which is the required solution.

Que) Solve $\left[\log(x^2+y^2) + \frac{2x^2}{(x^2+y^2)}\right]dx + \frac{2xy}{(x^2+y^2)}dy = 0$

$\Rightarrow$ on comparing with $Mdx + Ndy = 0$

$$M = \log(x^2+y^2) + \frac{2x^2}{(x^2+y^2)} \quad \text{and} \quad N = \frac{2xy}{(x^2+y^2)}$$

Diff. w.r.t. y partially

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(\log(x^2+y^2)\right) + 2x^2 \cdot \frac{\partial}{\partial y}\left(\frac{1}{x^2+y^2}\right)$$

$$\frac{\partial M}{\partial y} = \frac{1}{(x^2+y^2)}(0+2y) + 2x^2\left(\frac{-1}{(x^2+y^2)^2} \times (2y)\right)$$

$$\frac{\partial M}{\partial y} = \frac{2y}{x^2+y^2} - \frac{4x^2y}{(x^2+y^2)^2}$$

$$\frac{\partial M}{\partial y} = \frac{2y(x^2+y^2) - 4x^2y}{(x^2+y^2)^2}$$

$$\frac{\partial M}{\partial y} = \frac{2x^2y + 2y^3 - 4x^2y}{(x^2+y^2)^2} = \boxed{\frac{-2x^2y + 2y^3}{(x^2+y^2)^2}} \rightarrow (1)$$

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Now, $N = \frac{2xy}{x^2+y^2}$

Diff. w.r.t. x partially

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(\frac{2xy}{x^2+y^2}\right)$$

$$\frac{\partial N}{\partial x} = \frac{(x^2+y^2)\left(\frac{\partial}{\partial x}(2xy)\right) - 2xy\frac{\partial}{\partial x}(x^2+y^2)}{(x^2+y^2)^2}$$

$$\frac{\partial N}{\partial x} = \frac{(x^2+y^2)(2y) - 2xy(2x+0)}{(x^2+y^2)^2}$$

$$\therefore \frac{\partial N}{\partial x} = \frac{2x^2y + 2y^3 - 4x^2y}{(x^2+y^2)^2}$$

$$\boxed{\frac{\partial N}{\partial x} = \frac{-2x^2y + 2y^3}{(x^2+y^2)^2}} \longrightarrow (2)$$

from (1) & (2)

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ The given D.E. is exact

$\therefore$ the general solution is given by

$$\underset{(x=\text{const})}{\int N\,dy} + \int\begin{pmatrix}\text{The terms of } M\\ \text{not containing } y\end{pmatrix}dx = c$$

$$\int \frac{2xy}{(x^2+y^2)}\,dy + \int 0\,dx = c$$

$$x\int \frac{2y}{x^2+y^2}\,dy = c$$

$\int \frac{f'(x)}{f(x)}\,dx = \ln(f(x)) + c$

$$\boxed{x\ln(x^2+y^2) = c}$$

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* Non-Exact Differential Equations Reducible to Exact Form.

* Integrating Factor:- (I.F.) An integrating factor is a function of x and y, which when multiplied to given non-exact DE will make it exact.

$$\text{Non Exact DE} \xrightarrow[\text{by I.F.}]{\text{multiply it}} \text{Exact DE}$$

* Some Rules to find I.F.

① **Rule 1:-** If the D.E. is homogenous and $Mdx + Ndy \neq 0$ then $\text{I.F.} = \dfrac{1}{xM + yN}$

② **Rule 2:-** If DE is of the form $y f_1(x,y) + x f_2(x,y) = 0$ and $xM - yN \neq 0$ then $\text{I.F.} = \dfrac{1}{xM - yN}$

③ **Rule 3:-** It is applicable only for 'xdy'

IF $\dfrac{\left(\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}\right)}{N} = f(x)$ then $\text{I.F.} = e^{\int f(x)dx}$

④ **Rule 4:-** It is applicable only for 'ydx'.

If $\dfrac{\left(\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y}\right)}{M} = f(y)$ then $\text{I.F.} = e^{\int f(y)dy}$

*⑤ **Rule 5:-** If the given D.E. $Mdx + Ndy = 0$ can be written as

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$$x^a y^b (my\,dx + nx\,dy) + x^r y^s (py\,dx + qx\,dy) = 0$$

where a, b, m, n, r, s, p, q are constants having any value then

$$I.F. = x^h y^k$$

where h and k are constants to be determined

$$\therefore \quad nh - mk = (m-n) + (mb - na)$$
$$qh - pk = (p-q) + (ps - qr)$$

provided $mq - np \neq 0$.

Que) Solve $x(x-y)\dfrac{dy}{dx} = y(x+y)$

Rule (1)

$$\Rightarrow \quad (x^2 - xy)\,dy = (xy + y^2)\,dx$$
$$-(xy + y^2)\,dx + (x^2 - xy)\,dy = 0$$
$$\therefore \quad (-xy - y^2)\,dx + (x^2 - xy)\,dy = 0$$

comparing with

$$M\,dx + N\,dy = 0$$
$$M = -xy - y^2 \quad \text{and} \quad N = x^2 - xy$$

Diff w.r.t. y partially

$$\frac{\partial M}{\partial y} = -x(1) - 2y$$
$$\frac{\partial M}{\partial y} = -x - 2y$$

diff w.r.t. x partially

$$\frac{\partial N}{\partial x} = 2x - y$$
$$\frac{\partial N}{\partial x} = 2x - y$$

$$\therefore \quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

the given D.E. is non-exact

Now find I.F., As given D.E. is homogenous.

$$\left(\frac{dy}{dx} = \frac{xy + y^2}{x^2 - xy}\right)$$

By Rule (1)

$$I.F. = \frac{1}{xM + yN}$$

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$$I.F. = \frac{1}{x(-xy-y^2)+y(x^2-xy}$$

$$I.F. = \frac{1}{-\cancel{x^2y}-xy^2+\cancel{x^2y}-xy^2}$$

$$I.F. = \frac{-1}{2xy^2}$$

$\therefore$ let $I.F. [Mdx + Ndy] = 0$

$$\therefore \frac{-1}{2xy^2}\left[(-xy-y^2)\,dx + (x^2-xy)\,dy\right] = 0$$

$\therefore$ We can ignore the constant term $\frac{-1}{2}$ in $\frac{-1}{2xy^2}$ as it get multiply to 0 on R.H.S.

$$\therefore \frac{1}{xy^2}\left[(-xy-y^2)\,dx + (x^2-xy)\,dy\right] = 0$$

$$\therefore \left(\frac{-xy}{xy^2} - \frac{\cancel{y^2}}{x\cancel{y^2}}\right)dx + \left(\frac{x^2}{xy^2} - \frac{xy}{xy^2}\right)dy = 0$$

$\therefore$

$$\left(\frac{-1}{y} - \frac{1}{x}\right)dx + \left(\frac{x}{y^2} - \frac{1}{y}\right)dy = 0$$

Now this equation is exact and general solution is given by

$$\int_{y=const} M\,dx + \int [\text{The terms of } N \text{ not containing } x]\,dy = C$$

$$+\int_{y=const}\left(\frac{-1}{y}-\frac{1}{x}\right)dx + \int \frac{-1}{y}\,dy = C$$

$$\therefore -\int_{y=const}\frac{1}{y}\,dx - \int\frac{1}{x}\,dx - \int\frac{1}{y}\,dy = C$$

$$\therefore -\frac{1}{y}(x) - \log(x) - \log y = C$$

$$\therefore \boxed{\frac{x}{y} + \log(x) + \log y = c_1} \qquad \boxed{c_1 = -c}$$

Required solution.

Que) Solve $(1+xy)\,y\,dx + (1-xy)\,x\,dy = 0$

Rule② ⇒ Given:

$$(y + xy^2)\,dx + (x - x^2y)\,dy = 0 \longrightarrow ①$$

on comparing with $M\,dx + N\,dy = 0$

$$M = y + xy^2 \quad \text{and} \quad N = x - x^2y$$

diff w.r.t. y partially	diff w.r.t. x partially
$\frac{\partial M}{\partial y} = 1 + 2xy$	$\frac{\partial N}{\partial x} = 1 - 2xy$

$$\therefore \quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Given D.E. is not exact.

find I.F.

As given D.E. of the form.

$$y\,f_1(x,y)\,dx + x\,f_2(x,y)\,dy = 0$$

By Rule ②

$$I.F. = \frac{1}{xM - yN}$$

$$I.F = \frac{1}{x(y+xy^2) - y(x-x^2y)}$$

$$I.F. = \frac{1}{\cancel{xy} + x^2y^2 - \cancel{xy} + x^2y^2} = \frac{1}{2x^2y^2}$$

Ignoring the constant

$$I.F. = \frac{1}{x^2y^2}$$

Let $I.F. \,[M\,dx + N\,dy] = 0$

$$\frac{1}{x^2y^2}\left[(y+xy^2)\,dx + (x-x^2y)\,dy\right] = 0$$

$$\left(\frac{y}{x^2y^2} + \frac{xy^2}{x^2y^2}\right)dx + \left(\frac{x}{x^2y^2} - \frac{x^2y}{x^2y^2}\right)dy = 0$$

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$\left(\frac{1}{x^2y}+\frac{1}{x}\right)dx+\left(\frac{1}{xy^2}-\frac{1}{y}\right)dy=0$

Now the given D.E. is exact

$\therefore \int_{y=const.} M\,dx + \int \begin{pmatrix}\text{The terms of } N \\ \text{not containing } x\end{pmatrix} dy = c$

$\int_{y=const.} \left(\frac{1}{x^2y}+\frac{1}{x}\right)dx + \int\left(\frac{-1}{y}\right)dy = c$

$\therefore \frac{1}{y}\int \frac{1}{x^2}dx + \int \frac{1}{x}dx - \int \frac{1}{y}dy = c$

$\frac{1}{y}\left(\frac{-1}{x}\right) + \log(x) - \log(y) = c$

$\frac{-1}{xy} + \log\left(\frac{x}{y}\right) = c$

$\therefore \boxed{\log\left(\frac{x}{y}\right) - \frac{1}{xy} = c}$

Ex) $(x^2+y^2+1)\,dx - 2xy\,dy = 0$

Rule ③ ⇒ Given $(x^2+y^2+1)\,dx - 2xy\,dy = 0$

on comparing with $M\,dx + N\,dy = 0$

$M = x^2+y^2+1$ and $N = -2xy$

diff. w.r.t. y partially | Diff w.r.t. x partially

$\frac{\partial M}{\partial y} = 2y$ | $\frac{\partial N}{\partial x} = -2y$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ The given D.E. is not exact, find I.F.

Here Rule 1 and Rule 2 are not applicable.

$(x^2+y^2+1)\,dx - 2y\,\underline{(x\,dy)} = 0$

Rule ③ is applicable.

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$\therefore$ let, $\dfrac{\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)}{N} = \dfrac{(2y)-(-2y)}{-2xy}$

$= \dfrac{2y+2y}{-2xy}$

$= \dfrac{4y}{-2xy}$

$= \dfrac{-2}{x} = \boxed{f(x)}$

$I.F. = e^{\int f(x)dx} = e^{\int -\frac{2}{x}dx} = e^{-2\log x} = x^{-2}$

$\therefore \quad I.F. = \dfrac{1}{x^2}$

let $\quad I.F. \,[Mdx + Ndy] = 0$

$\dfrac{1}{x^2}\left[(x^2+y^2+1)\,dx + (-2xy)\,dy\right] = 0$

$\therefore \left(\dfrac{x^2}{x^2} + \dfrac{y^2}{x^2} + \dfrac{1}{x^2}\right)dx - \dfrac{2xy}{x^2}\,dy = 0$

$\left(1 + \dfrac{y^2}{x^2} + \dfrac{1}{x^2}\right)dx - \dfrac{2y}{x}\,dy = 0$

$\therefore$ Now the given D.E. is exact. The general solution is given by

$\int_{y=const} M dx + \int \left(\begin{array}{l}\text{The terms of } N \text{ not}\\ \text{containing } x\end{array}\right) dy = c$

$\int_{y=const} \left(1 + \dfrac{y^2}{x^2} + \dfrac{1}{x^2}\right)dx + \int 0\,dy = c$

$\therefore \int dx + y^2 \int \dfrac{dx}{x^2} + \int \dfrac{1}{x^2}dx = c$

$$x + y^2\left(\frac{-1}{x}\right) - \frac{1}{x} = C$$

$$\boxed{x - \frac{y^2}{x} - \frac{1}{x} = C}$$

This is required solution.

H.W. Ex ① Solve $(x^4e^x - 2mxy^2)\,dx + 2mx^2y\,dy = 0$

$\Rightarrow$ Rule ③, I.F. $= \frac{1}{x^4}$, $f(x) = \frac{-4}{x}$

Solution: $e^x + \frac{my^2}{x^2} = c$

② Solve $(2x\log x - xy)\,dy + 2y\,dx = 0$

$\Rightarrow$ Rule ③ $f(x) = \frac{-1}{x}$, I.F. $= \frac{1}{x}$

Solution: $2y\log x - \frac{y^2}{2} = c$

Ex) $y\log y\,dx + (x - \log y)\,dy = 0$

Rule④ $\Rightarrow$ Given $y\log y\,dx + (x - \log y)\,dy = 0$

on comparing with $M\,dx + N\,dy = 0$

$M = y\log y$ and $N = x - \log y$

diff w.r.t. y partially

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(y\log y)$$

$$= \cancel{y}\left(\frac{1}{\cancel{y}}\right) + \log y\,(1)$$

$$\frac{\partial M}{\partial y} = 1 + \log y$$

diff. w.r.t. x partially

$$\frac{\partial N}{\partial x} = 1 - 0$$

$$\frac{\partial N}{\partial x} = 1$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

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The given D.E. is non-exact, find I.F.

As $\log y\,(y\,dx) + (x-\log y)\,dy = 0$

By Rule (4),

$$\frac{\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)}{M} = \frac{1-(1+\log y)}{y\log y} = \frac{1-1-\log y}{y\log y}$$

$$= \frac{0-\cancel{\log y}}{y\,\cancel{\log y}}$$

$$= \frac{-1}{y}$$

$$I.F. = e^{\int f(y)dy} = e^{\int \frac{-1}{y}dy} = e^{-\log y} = e^{\log y^{(-1)}} = y^{-1} = \frac{1}{y}$$

$\therefore \boxed{I.F. = \frac{1}{y}}$

Let $I.F. \,[M dx + N dy] = 0$

$$\frac{1}{y}\left[y\log y\,dx + (x-\log y)\,dy\right] = 0$$

$$\therefore \frac{\cancel{y}\log y}{\cancel{y}}\,dx + \left(\frac{x}{y} - \frac{\log y}{y}\right)dy = 0$$

$\therefore$ Now the given D.E. is exact and general solution is given by

$$\int_{y=\text{const}} M dx + \int \left(\begin{matrix}\text{The terms of } N \text{ not}\\ \text{containing } x\end{matrix}\right) dy = C$$

$$\int_{y=\text{const}} \log y\,dx + \int -\frac{\log y}{y}\,dy = C$$

$$\log y \int dx - \int \frac{\log y}{y}\,dy = C$$

put $t = \log y$

$\therefore dt = \frac{1}{y} dy$

Above eqn becomes

$$\log y \int dx - \int t \, dt = c$$

$$\log y \, (x) - \frac{t^2}{2} = c$$

but $t = \log y$

$$\therefore \boxed{x \log y - \frac{(\log y)^2}{2} = c}$$

Ex① Solve $y(2x^2 y + e^x) dx = (e^x + y^3) dy$

H.W. $\Rightarrow$ $f(y) = -\frac{2}{y}$, $I.F. = \frac{1}{y^2}$ and soln, $\frac{2x^3}{3} + \frac{e^x}{y} - \frac{y^2}{2} = c$

Ex) Solve $(x^2 y + y^4) dx + (2x^3 + 4xy^3) dy = 0$

Rule5 $\Rightarrow$ Given, $(x^2 y + y^4) dx + (2x^3 + 4xy^3) dy = 0$
on comparing with $Mdx + Ndy = 0$

$M = x^2 y + y^4$ and $N = 2x^3 + 4xy^3$

diff. w.r.t. y partially	diff. w.r.t. x partially
$\frac{\partial M}{\partial y} = x^2(1) + 4y^3$	$\frac{\partial N}{\partial x} = 2(3x^2) + 4y^3(1)$
$\frac{\partial M}{\partial y} = x^2 + 4y^3$	$\frac{\partial N}{\partial x} = 6x^2 + 4y^3$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ The given D.E. is not exact.
Now the given D.E. can be written as

$$x^2 y \, dx + y^4 dx + 2x^3 dy + 4xy^3 dy = 0$$

$$(x^2 y \, dx + 2x^3 dy) + (y^4 dx + 4xy^3 dy) = 0$$

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$x^2(ydx+2xdy)+y^3(ydx+4xdy)=0$

$\therefore x^2y^0(ydx+2xdy)+x^0y^3(ydx+4xdy)=0$

The given D.E is of the form

$x^ay^b(mydx+nxdy)+x^ry^s(pydx+qxdy)=0$

where

$a=2, b=0, m=1, n=2$ / $r=0, S=3, P=1, q=4$

and $\boxed{I.F. = x^h y^k}$

where h and k are constants to be determined

We have

$nh-mk=(m-n)+(mb-na)$

$2h-K=(1-2)+(1(0)-2(2))$

$2h-K=-1+(0-4)$

$\boxed{2h-K=-5} \longrightarrow ①$

Also

$qh-pk=(p-q)+(ps-qr)$

$4h-K=(1-4)+(1(3)-4(0))$

$4h-K=-3+3-0$

$\boxed{4h-K=0} \longrightarrow ②$

subtract eqn ① from eqn ②

$4h-K=0 \longrightarrow ②$

$-2h \mp K = \mp 5 \longrightarrow ①$

$2h=5$

$\boxed{h=\frac{5}{2}}$

put $h=\frac{5}{2}$ in eqn ②

$4\left(\frac{5}{2}\right)-K=0$

$\boxed{K=10}$

$\therefore \boxed{I.F. = x^{\frac{5}{2}}y^{10}}$.

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Let I.F. $[Mdx + Ndy] = 0$

$$x^{5/2} y^{10} \left\{ [x^2 y + y^4] dx + [2x^3 + 4xy^3] dy \right\} = 0$$

$$\therefore x^{5/2} y^{10} x^2 y \, dx + x^{5/2} y^{10} y^4 \, dx + (2x^3 x^{5/2} y^{10} + 4xy^3 \times x^{5/2} y^{10}) dy = 0$$

$(\because a^m \times a^n = a^{m+n})$

$$\therefore (x^{9/2} y^{11} + x^{5/2} y^{14}) dx + (2x^{11/2} y^{10} + 4x^{7/2} y^{13}) dy = 0$$

$\therefore$ Now the given D.E. is exact

The general solution is given by,

$$\int_{y=const} M dx + \int (\text{The terms of } N \text{ not containing } x) dy = c$$

$$\int_{y=const} (x^{9/2} y^{11} + x^{5/2} y^{14}) dx + \int 0 \, dy = C$$

$$\therefore y^{11} \int x^{9/2} dx + y^{14} \int x^{5/2} dx = c$$

$$y^{11} \left(\frac{x^{11/2}}{(11/2)} \right) + y^{14} \left(\frac{x^{7/2}}{7/2} \right) = C$$

$$\therefore \boxed{\frac{2x^{11/2} y^{11}}{11} + \frac{2x^{7/2} y^{14}}{7} = C}$$

which is a required solution.

* Transformation to polar co-ordinates

D.E. of 1st order and 1st degree $Mdx + Ndy = 0$

↓

Convert the D.E. using $x^2+y^2 = r^2$, $x = r\cos\theta$, $y = r\sin\theta$, $\theta = \tan^{-1}(y/x)$, $xdx + ydy = rdr$ and $xdx - ydy = r^2 d\theta$.

↓

D.E. gets converted to polar coordinate. Use V.S. forms.

↓ V.S. = Variable seprable form

Resubstitute the values of r and θ

Ex) Solve $(xdy - ydx) + \dfrac{1}{(x^2+y^2)}(xdx + ydy) = 0$

⇒ Transforming the given D.E. to polar co-ordinates by using

$x = r\cos\theta$, $y = r\sin\theta$, $\theta = \tan^{-1}(y/x)$,

$x^2+y^2 = r^2$, $xdx + ydy = rdr$ and

$xdy - ydx = r^2 d\theta$.

$$\therefore \quad r^2 d\theta + \frac{1}{r^2}(rdr) = 0$$

$$r^2 d\theta = -\frac{1}{r}dr$$

$$d\theta = \frac{-1}{r^3}dr \qquad \text{(Variable seprable form)}$$

on integrating both sides.

$$\int d\theta = -\int \frac{1}{r^3}\, dr$$

$$\theta = -\left[-\frac{1}{2r^2}\right] + C$$

$$\theta = \frac{1}{2r^2} + C$$

Resubstituting $\theta = \tan^{-1}(y/x)$ & $r^2 = x^2+y^2$

$$\therefore \boxed{\tan^{-1}(y/x) = \frac{1}{2(x^2+y^2)} + C}$$

Ex) Solve $x\,dy - y\,dx = (x^2+y^2)(x\,dx + y\,dy)$

H.W. ⇒ Soln:- $\tan^{-1}\left(\frac{y}{x}\right) = \frac{x^2+y^2}{2} + C.$

* Flow chart *

D.E. of 1st order and 1st degree $M\,dx + N\,dy = 0$

↓

IF. $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ then DE is exact

↓

Solution of exact DE is $\underset{y = \text{const.}}{\int M\,dx} + \int \left(\text{the terms of } N \text{ not containing } x\right) dy = C$

* Linear Differential Equations :-

* L.D.E. :- The differential equation is linear if the dependant variable (y) and it's derivative $\left(\frac{dy}{dx}\right)$ occurs only in first degree and are not multiplied together.

Ex) ① $x\frac{dy}{dx} + 2y = 4x^2$ is linear D.E.

② $y\frac{dy}{dx} - 4 = x$ is not linear D.E.

because dependant variable (y) and it's derivative are multiplied together.

* The general forms of a linear DE :-

Ⓐ $$\frac{dy}{dx} + Py = Q.$$

Single terms of y, linear in y.

where P and Q are functions of x or constants.

$$\therefore \quad I.F. = e^{\int P\,dx}$$

and general solution is

$$y\,(I.F.) = \int Q\,(I.F)\,dx + c$$

Ⓑ $$\frac{dx}{dy} + Px = Q$$

single term of x, linear in x.

$\therefore \quad I.F = e^{\int P\,dy}$ and general solution

$$x\,(I.F) = \int Q\,(I.F)\,dy + c$$

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Ex) $(x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$

Type(1)

$\Rightarrow (x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2} \longrightarrow (1)$

There is only one term of y, therefore the given D.E. is linear in y. Also for linear D.E. coefficient of $\frac{dy}{dx}$ must be one.

$\therefore$ Divide eqn (1) by (x^2+1) throughout

$$\frac{dy}{dx} + \left(\frac{4x}{(x^2+1)}\right)y = \frac{1}{(x^2+1)^3}$$

compare with $\frac{dy}{dx} + Py = Q$

$\therefore P = \frac{4x}{(x^2+1)}$ and $Q = \frac{1}{(x^2+1)^3}$

$$I.F. = e^{\int P dx}$$

$$I.F. = e^{\int\left(\frac{4x}{x^2+1}\right)dx}$$

$$I.F. = e^{2\left(\int\frac{2x}{x^2+1}dx\right)}$$

$$I.F. = e^{2\log(x^2+1)} \qquad \left(\int\frac{f'(x)}{f(x)}dx = \log(f(x))\right)$$

$$I.F. = e^{\log(x^2+1)^2} \qquad \left(\log m^n = n \times \log m\right)$$

$$I.F. = (x^2+1)^2$$

$\therefore$ The general solution is given by

$$y(I.F.) = \int Q(I.F.)\,dx + C$$

$$y(x^2+1)^2 = \int\frac{1}{(x^2+1)^3}(x^2+1)^2 dx + C$$

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$$y(x^2+1)^2 = \int \frac{1}{x^2+1}\,dx + c$$

$$\therefore \boxed{y(x^2+1)^2 = \tan^{-1}(x) + c}$$

Ex) Type ①

$$\frac{dy}{dx} + y\cot x = \sin 2x$$

$\Rightarrow$ Given $\frac{dy}{dx} + y\cot x = \sin 2x \longrightarrow ①$

on comparing with $\frac{dy}{dx} + Py = Q$

$P = \cot x$ and $Q = \sin 2x$

$$\therefore I.F. = e^{\int P dx}$$

$$= e^{\int \cot x\,dx}$$

$$= e^{\log(\sin x)}$$

$$\boxed{I.F. = \sin x}$$

The general solution is given by

$$y(I.F.) = \int Q\,(I.F.)\,dx + c$$

$$y\sin x = \int \sin 2x \cdot \sin x\,dx + c$$

$$y\sin x = \int 2\sin x\cos x \cdot \sin x\,dx + c$$

$(\sin 2\theta = 2\sin\theta\cos\theta)$

$$y\sin x = 2\int \sin^2 x \cdot \cos x\,dx + c$$

$$\therefore y\sin x = 2\,\frac{\sin^{(2+1)} x}{2+1} + c \qquad \left(\int [f(x)]^n \cdot f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1}\right)$$

$$y\sin x = \frac{2}{3}\sin^3 x + c$$

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HW Ex) Solve $x\cos x \frac{dy}{dx} + (\cos x - x\sin x)y = 1$

$\Rightarrow$ I.F. $= x\cos x$ and soln $xy\cos x = x + c$

Ex) Solve $\cos x \frac{dy}{dx} + y = \sin x$

Type ①

$\Rightarrow$ Divide by $\cos x$ throughout

$$\frac{dy}{dx} + \left(\frac{1}{\cos x}\right)y = \frac{\sin x}{\cos x}$$

but $\frac{1}{\cos x} = \sec x$ and $\frac{\sin x}{\cos x} = \tan x$

$$\therefore \frac{dy}{dx} + (\sec x)y = \tan x$$

compare with $\frac{dy}{dx} + Py = Q$

$\therefore P = \sec x$ and $Q = \tan x$

$$I.F. = e^{\int P dx} = e^{\int \sec x dx} = e^{\log(\sec x + \tan x)} = \sec x + \tan x$$

$\therefore$ I.F. $= (\sec x + \tan x)$

The general solution is given by

$$y(I.F) = \int Q(I.F)\,dx + c$$

$$y(\sec x + \tan x) = \int \tan x \times (\sec x + \tan x)\,dx + c$$

$$\therefore y(\sec x + \tan x) = \int \sec x \tan x\,dx + \int \tan^2 x\,dx + c$$

$$\therefore y(\sec x + \tan x) = \sec x + \int (\sec^2 x - 1)\,dx + c$$

$\left(\because \int \sec x \tan x\,dx = \sec x \text{ and } 1 + \tan^2 x = \sec^2 x\right)$

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$\therefore y(\sec x+\tan x) = \sec x + \int \sec^2 x\,dx - \int 1\,dx + c$

$$\boxed{y(\sec x+\tan x) = \sec x + \tan x - x + c}$$

which is a required solution.

Ex) Solve $(1+y^2) + \left(x - e^{\tan^{-1}y}\right)\dfrac{dy}{dx} = 0$

H.W.

Type ② ⇒ $I.F. = e^{\tan^{-1}y}$ and soln $xe^{\tan^{-1}y} = \tan^{-1}y + c$

Ex) Solve $y\log y\,dx + (x-\log y)\,dy = 0$

Type ② ⇒ Given $y\log y\,dx + (x-\log y)\,dy = 0$

As there is only one term of x, arrange the given DE as $\dfrac{dx}{dy}$

$\therefore y\log y\,dx = -(x-\log y)\,dy$

$\therefore y\log y\,\dfrac{dx}{dy} = -x + \log y$

$\therefore y\log y\,\dfrac{dx}{dy} + x = \log y$

divide by 'ylogy' through out

$$\frac{dx}{dy} + \left(\frac{1}{y\log y}\right)x = \frac{\cancel{\log y}}{y\cancel{\log y}}$$

$$\therefore \frac{dx}{dy} + \left(\frac{1}{y\log y}\right)x = \frac{1}{y}$$

on comparing with $\dfrac{dx}{dy} + Px = Q$

$$P = \frac{1}{y\log y} \quad \text{and} \quad Q = \frac{1}{y}$$

$$I.F. = e^{\int P\,dy} = e^{\int \frac{1}{y\log y}dy} = e^{\log t} = t = \log y$$

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$\because$ put $\log y = t$

$\frac{1}{y}dy = dt$

$\Rightarrow e^{\int \frac{1}{y\log y}dy} = e^{\int \frac{1}{t}dt} = e^{\log t} = t.$

$\therefore$ I.F. $= \log y$

The general solution is given by

$$x(I.F.) = \int Q\,(I.F.)\,dy + c$$

$$\therefore x\log y = \int \frac{1}{y}\log y\,dy + c$$

put $\log y = t$

$\frac{1}{y}dy = dt$

$$x\log y = \int t\,dt + c$$

$$x\log y = \frac{t^2}{2} + c$$

$$\therefore x\log y = \frac{(\log y)^2}{2} + c$$

$$\therefore \boxed{x\log y = \frac{1}{2}(\log y)^2 + c}$$

Ex) (1) Solve $\frac{dy}{dx} + (1+2x)y = e^{-x^2}$

H.W.

$\Rightarrow$ I.F. $= e^{(x+x^2)}$ and sol^n. $ye^{(x+x^2)} = e^x + c$

(2) Solve, $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right)\frac{dx}{dy} = 1$

$\Rightarrow$ I.F. $= e^{2\sqrt{x}}$ and sol^n $ye^{2\sqrt{x}} = 2\sqrt{x} + c$

Flow chart for L.D.E.

1st order and 1st degree DE

$M\,dx + N\,dy = 0$

linear in y

If the DE contains single term of y then arrange as $\frac{dy}{dx} + Py = Q$

$I.F. = e^{\int P\,dx}$

The general solution is

$y(I.F.) = \int Q(I.F.)\,dx + c$

linear in x

If the DE contains single term of x then arrange as $\frac{dx}{dy} + Px = Q$

$I.F. = e^{\int P\,dy}$

The general solution is

$y(I.F.) = \int Q(I.F.)\,dy + c$

* Bernoulli's Differential Equation *

* Equations Reducible to L.D.E.:-

The D.E. of the form

$$\frac{dy}{dx} + Py = Qy^n \quad \text{or} \quad \frac{dx}{dy} + Px = Qx^n$$

are called as Bernoulli's Differential equations

(A) $\frac{dy}{dx} + Py = Qy^n$

Divide by y^n throughout

$$\frac{1}{y^n}\frac{dy}{dx} + Py^{1-n} = Q$$

put $y^{1-n} = u$

Diff. w.r.t. x then ean get converted into

$$\frac{du}{dx} + Pu = Q$$

Which is linear DE in u.

(B) $\frac{dx}{dy} + Px = Qx^n$

divide by x^n throughout

$$x^{-n}\frac{dx}{dy} + Px^{1-n} = Q$$

put $x^{1-n} = u$

Diff. w.r.t. y

Then ean get converted into

$$\frac{du}{dy} + Pu = Q$$

Which is linear in u.

* **Note:** Also LDE of the form $f'(y)\frac{dy}{dx} + Pf(y) = Q$ and $f'(x)\frac{dx}{dy} + Pf(x) = Q$ can also be converted into LDE by putting $f(y) = u$ and $f(x) = u$ respectively

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Que: Solve $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

Type①

$\Rightarrow$ multiply by '−' sign throughout

$$\frac{dy}{dx} - xy = -y^3 e^{-x^2} \quad \text{... Bernoulli's eqn}$$

divide by y^3 throughout

$$y^{-3}\frac{dy}{dx} - xy^{-2} = -e^{-x^2} \longrightarrow ① \quad \left(\because \frac{1}{x^n} = x^{-n}\right)$$

substitute $y^{-2} = u$

Diff w.r.t. x

$$-2y^{-3}\frac{dy}{dx} = \frac{du}{dx}$$

$$y^{-3}\frac{dy}{dx} = \frac{-1}{2}\frac{du}{dx}$$

$\therefore$ eqn ① becomes

$$\frac{-1}{2}\frac{du}{dx} - xu = -e^{-x^2}$$

multiply by −2 throughout

$$\frac{du}{dx} + 2xu = 2e^{-x^2}$$

which is linear D.E. in u.

compare with $\frac{du}{dx} + Pu = Q$

$$P = 2x \text{ and } Q = 2e^{-x^2}$$

$$\therefore I.F. = e^{\int P dx}$$

$$= e^{\int 2x dx} = e^{\cancel{2}\left(\frac{x^2}{\cancel{2}}\right)} = e^{x^2}$$

The general solution is given by

$$u(I.F.) = \int Q(I.F.) dx + C$$

As eqn is linear in u, the general soln will start with u.

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$$ue^{x^2} = \int 2 e^{-x^2} x e^{x^2} dx + C$$

$$ue^{x^2} = \int 2\, dx + C$$

$$ue^{x^2} = 2x + C$$

but $u = y^{-2}$

$$y^{-2} e^{x^2} = 2x + C$$

$$\boxed{\frac{e^{x^2}}{y^2} = 2x + C}$$

Which is required solution.

Ex) $\frac{dy}{dx} - y \tan x = y^4 \sec x$

$\Rightarrow$ $\frac{dy}{dx} - (\tan x) y = y^4 \sec x \longrightarrow (1)$

Bernoulli's Equation

divide by y^4 throughout

$$\frac{1}{y^4}\frac{dy}{dx} - \tan x \left(\frac{y}{y^4}\right) = \sec x.$$

$$y^{-4}\frac{dy}{dx} - \tan x\, y^{-3} = \sec x \longrightarrow (2)$$

put $y^{-3} = u$

diff. w.r.t. x

$$-3y^{-4}\frac{dy}{dx} = \frac{du}{dx}$$

$$y^{-4}\frac{dy}{dx} = \frac{-1}{3}\frac{du}{dx}$$

$\therefore$ eqn (2) becomes

$$\frac{-1}{3}\frac{du}{dx} - \tan x\, u = \sec x$$

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Multiply by -3 throughout

$\therefore \frac{du}{dx} + 3\tan x\, u = -3\sec x$.

which is linear in u.

on comparing with $\frac{du}{dx} + Pu = Q$.

$P = 3\tan x$ and $Q = -3\sec x$.

$I.F. = e^{\int P dx}$

$I.F. = e^{\int 3\tan x\, dx} = e^{3\log(\sec x)} = e^{\log(\sec x)^3} = \sec^3 x$.

$\therefore I.F. = \sec^3 x$

the general solution is given by

$u(I.F.) = \int Q (I.F.) dx + c$

$u\sec^3 x = \int -3\sec x \cdot \sec^3 x\, dx + c$

$u\sec^3 x = -3 \int \sec^2 x \cdot \sec^2 x\, dx + c$

but $y^{-3} = u$ $\quad (\because 1+\tan^2 x = \sec^2 x)$

$\therefore y^{-3}\sec^3 x = -3 \int (1+\tan^2 x) \sec^2 x\, dx + c$

$\frac{\sec^3 x}{y^3} = -3\left[\int \sec^2 x\, dx + \int \tan^2 x \sec^2 x\, dx + c\right.$

put $t = \tan x$

$dt = \sec^2 x\, dx$

$\therefore \frac{\sec^3 x}{y^3} = -3\tan x + (-3)\int t^2\, dt + c$

$\frac{\sec^3 x}{y^3} = -3\tan x - \cancel{3}\left(\frac{t^3}{\cancel{3}}\right) + c$

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$$\boxed{\frac{\sec^3 x}{y^3} = -3\tan x - \tan^3 x + c}$$

Which is a required solution.

Ex) $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$

Type ③

$\Rightarrow$ Given $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y \longrightarrow (1)$

divide by $\cos^2 y$ throughout

$$\frac{1}{\cos^2 y}\frac{dy}{dx} + \frac{2x\sin y\cos y}{\cos^2 y} = x^3$$

$(\because \sin 2\theta = 2\sin\theta\cos\theta)$

$\therefore \sec^2 y\frac{dy}{dx} + (2\tan y)x = x^3 \longrightarrow (2)$

put $\tan y = u$

$\sec^2 y\frac{dy}{dx} = \frac{du}{dx}$

$\therefore$ eqn (2) becomes

$\therefore \frac{du}{dx} + 2xu = x^3 \longrightarrow (3)$

Which is linear in u.

$\therefore$ comparing with $\frac{du}{dx} + Pu = Q$

$P = 2x$ and $Q = x^3$

$$I.F. = e^{\int P dx} = e^{\int 2x dx} = e^{\cancel{2}\left(\frac{x^2}{\cancel{2}}\right)} = e^{x^2}$$

The General solution is given by

$u(I.F.) = \int Q(I.F.)dx + c$

$ue^{x^2} = \int x^3 e^{x^2} dx + c$

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$$ue^{x^2} = \int x^2 e^{x^2} (x\,dx) + c$$

put $x^2 = t$

$2x\,dx = dt$

$x\,dx = \frac{dt}{2}$

$$\therefore ue^{x^2} = \int \underset{u}{t}\,\underset{v}{e^t} \left(\frac{dt}{2}\right) + c$$

$$\int uv = u\int v - \int\left(\frac{d}{dx}(u)\int v\,dx\right)dx$$

$$ue^{x^2} = \frac{1}{2}\left[t\int e^t dt - \int\left(\frac{d}{dt}(t)\int e^t dt\right)dt\right] + c$$

$$ue^{x^2} = \frac{1}{2}\left[te^t - \int 1\times e^t dt\right] + c$$

$$ue^{x^2} = \frac{1}{2}\left[te^t - e^t\right] + c$$

but $t = e^{x^2}$

$$\therefore ue^{x^2} = \frac{1}{2}\left(x^2e^{x^2} - e^{x^2}\right) + c$$

Also $u = \tan y$.

$$\boxed{\tan y \cdot e^{x^2} = \frac{1}{2}e^{x^2}(x^2 - 1) + c}$$

which is required solution.

Que: Solve $\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^2 x$

H.W.

$\Rightarrow$ substitution $u = \sec y$, I.F. $= \sec x$.

Soln, $\sec y \cdot \sec x = \sin x + c$.